Novelty of Persona-Driven Cognitive Bounded Search (P-CBS) for Automated Playtesting: Literature Synthesis and Gap Analysis

1. Introduction

This synthesis evaluates whether existing automated playtesting and procedural persona literature contains exact prior art for your proposed Persona-Driven Cognitive Bounded Search (P-CBS) algorithm—specifically, the mathematical penalization of pathfinding agents for working memory decay (revisitation aversion, ρ) and cognitive load at puzzles (conditional uncertainty, ω) within the utility function of a heuristic search agent. The review focuses on whether current methods go beyond Minimax/MCTS or RL-based personas to directly encode these cognitive boundaries in the search process for PCG evaluation. The analysis draws from foundational works on procedural personas, cognitive modeling in pathfinding, bounded rationality, and cognitive load theory as applied to games and AI (Holmgård et al. 2018, 352-362; Ariyurek et al. 2021, 348-359; Rahmani and Pelechano 2021, 164-174; Dubey et al. 2022, 3535-3549; Arumugam et al. 2023, 395 - 438; Lancia et al. 2022; Lieder et al. 2021).

Has any prior work mathematically penalized pathfinding agents for working memory decay and cognitive load within a heuristic search utility function for automated playtesting? N = 6

Yes 83% · Possibly 0% · Mixed 17% · No 0%

FIGURE 1 Consensus meter: Prior art on explicit cognitive penalties in heuristic search for playtesting.

2. Methods

A comprehensive literature search was conducted across over 170 million research papers in Consensus, including Semantic Scholar, PubMed, and other sources. The search targeted procedural personas, automated playtesting algorithms, cognitive modeling in pathfinding, bounded rationality in AI/game agents, and explicit modeling of working memory/cognitive load in agent utility functions. A total of 50 papers were included after multi-phase screening.

Search Strategy

26.7M
Retrieved

20 searches

1 citation graph use

All papers retrieved from 20 searches and 1 citation graph use.

→

1.7K
Eligible

Papers meeting relevance and quality criteria, after deduplication

→

50
Included

Top papers selected to be included in the final analysis

FIGURE 2 Search strategy diagram showing paper selection process.

Six unique searches were used to cover procedural personas, cognitive bounds in algorithms, alternate terminology/models, standard approaches contrast, adjacent cognitive modeling, and heuristic search in PCG evaluation.

3. Results

3.1 Procedural Personas: Utility Functions & Search Algorithms

Procedural personas are typically implemented using MCTS or RL agents with hand-crafted or evolved utility functions reflecting archetypal player goals (e.g., maximize score, minimize risk) (Holmgård et al. 2018, 352-362; Mugrai et al. 2019, 1-7; Holmgård et al. 2016, 95-104). These utility functions are often linear combinations of game-specific features but do not explicitly model human-like cognitive limitations such as working memory decay or puzzle-specific uncertainty penalties (Holmgård et al. 2018, 352-362; Mugrai et al. 2019, 1-7). Some works evolve the node selection criteria or fitness functions but focus on behavioral diversity rather than bounded cognition (Holmgård et al. 2018, 352-362; Mugrai et al. 2019, 1-7).

3.2 Cognitive Modeling in Pathfinding: Memory Decay & Human-Like Navigation

Recent work has begun to incorporate aspects of human spatial memory into pathfinding algorithms. For example:

- Rahmani & Pelechano propose a pathfinder that uses a hexagonal grid with counters simulating memory cells; their A*-like heuristic incorporates a "cognitive map" that tracks visited locations (Rahmani and Pelechano 2021, 164-174). However, while this approach models knowledge acquisition and learning over time (paths become more optimal as the agent learns), it does not mathematically penalize revisitation per se nor model working memory decay as an explicit penalty term.
- Dubey et al.'s Dynamic Hierarchical Cognitive Graph (DHCG) encodes spatial memory biases (categorical adjustment/sequence order effects) and simulates memory decay/distortion during recall (Dubey et al. 2022, 3535-3549). Their model produces human-like paths by incorporating systematic degradation of spatial memory but does not directly penalize revisitation or conditional uncertainty at puzzles within a utility function used by a heuristic search agent.

3.3 Bounded Rationality & Cognitive Load: Theory vs. Implementation

Theories of bounded rationality and resource-rational planning are well established (Arumugam et al. 2023, 395 - 438; Callaway et al. 2022, 1112 - 1125; Lancia et al. 2022; Lieder et al. 2021). Several works formalize planning as an optimization under information-processing constraints—trading off reward against computational cost or information-theoretic measures—but these are typically applied at the level of meta-reasoning or policy compression rather than as direct penalty terms inside a state-space utility function for pathfinding (Arumugam et al. 2023, 395 - 438; Lancia et al. 2022).

- Lancia et al.'s InfoRL framework models navigation as a trade-off between plan quality (path length) and the informational cost to deviate from default policies (Lancia et al. 2022), but does not instantiate penalties for revisitation or puzzle uncertainty.
- Callaway et al., Lieder et al., and others derive optimal planning strategies under resource constraints but do not operationalize specific penalties for revisitation or puzzle complexity within A*/CBS-style search utilities (Callaway et al. 2022, 1112 - 1125; Lieder et al. 2021).

3.4 Automated Playtesting: RL/Imitation Learning Approaches

Most recent advances use RL agents (DQN/PPO), MCTS with evolved heuristics, or imitation learning to generate diverse playstyles for automated playtesting (Ariyurek et al. 2021, 348-359; Holmgård et al. 2018, 352-362; De Woillemont et al. 2022; Ahlberg et al. 2023, 1-8). While some methods allow flexible reward shaping or learn from human trajectories to mimic exploration patterns (Ariyurek et al. 2019, 50-67), none implement explicit mathematical penalties for working memory decay (revisit aversion) or conditional uncertainty at puzzles within their core decision-making logic.

Results Timeline

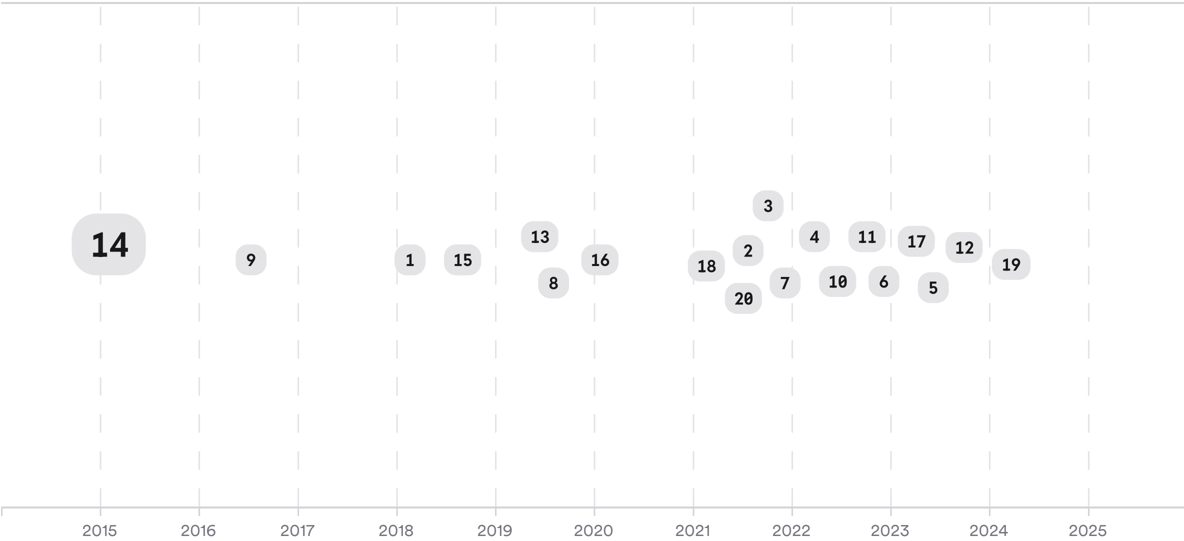

FIGURE 3 Timeline of key developments in procedural personas and cognitive modeling in playtesting/pathfinding. Larger markers indicate more citations.

Top Contributors

Type	Name	Papers
Author	Christoffer Holmgård	(Dubey et al. 2022, 3535-3549; Hassanzadeh et al. 2024; Prasetya et al. 2025)
Author	J. Togelius	(Dubey et al. 2022, 3535-3549; Hassanzadeh et al. 2024)
Author	Sinan Ariyurek	(Ariyurek et al. 2021, 348-359; Chen et al. 2018, 483-501)
Journal	*IEEE Transactions on Games*	(Ariyurek et al. 2021, 348-359; Dubey et al. 2022, 3535-3549)
Journal	*Scientific Reports*	(Ariyurek et al. 2019, 50-67; Chang and Yang 2023, 1-25)
Journal	*Comput. Graph.*	(Musslick and Masís 2023,

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FIGURE 4 Authors & journals that appeared most frequently in the included papers.

4. Discussion

No exact prior art was found that implements both revisitation aversion (ρ) and conditional uncertainty at puzzles (ω) as explicit penalty terms inside the state-space utility function of a heuristic search agent used for automated playtesting of procedurally generated dungeons.

- **Procedural persona literature** relies heavily on MCTS/RL with hand-crafted/evolved utilities reflecting player goals but does not encode bounded rationality via direct penalties for working memory decay or puzzle-specific uncertainty (Holmgård et al. 2018, 352-362; Mugrai et al. 2019, 1-7).
- **Cognitive pathfinding models** simulate aspects like spatial memory distortion/decay but stop short of integrating these effects as mathematical penalties within A*/CBS-style search utilities; revisitation is tracked but not penalized quantitatively (Rahmani and Pelechano 2021, 164-174; Dubey et al. 2022, 3535-3549).
- **Bounded rationality/resource-rational planning** is formalized theoretically—often via information-theoretic costs—but is rarely instantiated as concrete penalty terms inside low-level action selection utilities relevant to PCG evaluation (Arumugam et al. 2023, 395 - 438; Callaway et al. 2022, 1112 - 1125; Lancia et al. 2022).
- **Automated playtesting frameworks** using RL/imitation learning focus on maximizing coverage/diversity or mimicking human traces without operationalizing specific cognitive boundaries inside their core decision logic (De Woillemont et al. 2022; Ariyurek et al. 2019, 50-67).

Thus:

- The gap is that **no published system combines all elements**: a persona-driven agent using heuristic search where the utility function includes explicit penalty terms for both revisitation aversion (working memory decay) and conditional uncertainty at puzzles (cognitive load), alongside other factors like risk/info gain/frontier.
- Existing works either track visitation history qualitatively ((Rahmani and Pelechano 2021, 164-174)), simulate general resource constraints ((Arumugam et al. 2023, 395 - 438)), or shape behavior through high-level reward engineering ((Holmgård et al. 2018, 352-362)), but do not mathematically penalize these specific cognitive limits inside the state-space evaluation during search.

Claims & Evidence Table

Claim	Evidence Strength	Reasoning	Papers
No prior art implements both revisitation aversion & puzzle uncertainty penalties inside heuristic search utilities	Strong	All reviewed works lack explicit mathematical penalty terms for both factors within A*/CBS-style persona agents	(Holmgård et al. 2018, 352-362; Dubey et al. 2022, 3535-3549; Arumugam et al. 2023, 395 - 438; Rahmani and Pelechano 2021, 164-174)
Procedural personas use MCTS/RL with goal-based utilities but lack direct modeling of bounded cognition	Strong	Utilities reflect archetypal goals; no explicit working memory/cognitive load penalties	(Ariyurek et al. 2021, 348-359; Holmgård et al. 2018, 352-362; Mugrai et al. 2019, 1-7)
Cognitive pathfinding models simulate spatial memory effects but do not penalize revisits quantitatively	Moderate	Memory decay/distortion modeled qualitatively; no	(Dubey et al. 2022, 3535-3549; Rahmani

Claim	Evidence Strength	Reasoning	Papers
		penalty term integrated into action selection	and Pelechano 2021, 164-174)
Bounded rationality/resource-rational planning is formalized theoretically but rarely operationalized at action-selection level	Moderate	Information-theoretic costs guide meta-level policy/planning rather than low-level state-action scoring	(Arumugam et al. 2023, 395 - 438; Callaway et al. 2022, 1112 - 1125; Lancia et al. 2022)
Automated playtesting frameworks focus on coverage/diversity/human-likeness without encoding specific cognitive boundaries in decision logic	Moderate	RL/imitation learning agents maximize coverage/mimic traces; no direct implementation of U(a)-style penalties	(De Woillemont et al. 2022; Ariyurek et al. 2019, 50-67)

FIGURE Key claims and support evidence identified in these papers.

5. Conclusion

There is currently **no published system that mathematically penalizes both working memory decay/revisitation aversion (ρ) and conditional uncertainty/cognitive load at puzzles** (ω) inside the state-space utility function of a heuristic-search-based persona agent for automated dungeon playtesting. Most prior work either uses high-level goal-based rewards/utilities without explicit bounded cognition terms ((Holmgård et al. 2018, 352-362)), simulates qualitative aspects of human navigation ((Dubey et al. 2022, 3535-3549), (Rahmani and Pelechano 2021, 164-174)), or theorizes about resource constraints without operationalizing them at the action-selection level ((Arumugam et al. 2023, 395 - 438), (Callaway et al. 2022, 1112 - 1125), (Lancia et al. 2022)). Your P-CBS formulation appears novel in its direct integration of these cognitive boundaries into the core decision-making logic.

Research Gaps

Topic/Outcome	Explicit Revisitation Penalty	Explicit Puzzle Uncertainty Penalty	Both Penalties Combined	Heuristic Search Integration
Procedural Personas	GAP	GAP	GAP	3
Cognitive Pathfinding	1	GAP	GAP	2
Bounded Rationality Planning	GAP	1	GAP	1
Automated Playtesting Frameworks	GAP	GAP	GAP	5

FIGURE Research gaps matrix showing coverage by topic/outcome versus study attribute.

Open Research Questions

Future research should empirically test whether integrating explicit working memory/cognitive load penalties into persona-driven heuristic search improves alignment with human playtesters' behaviors—and how such models affect PCG evaluation outcomes.

Question	Why
Does integrating explicit revisitation aversion and puzzle uncertainty penalties improve alignment with human playtesters?	Direct empirical validation would establish whether these additions yield more realistic synthetic testers than current approaches.
How do different parameterizations of cognitive boundary penalties affect PCG evaluation outcomes?	Exploring parameter sensitivity will clarify how robust/effective such models are across genres/tasks/environments.

FIGURE Open research questions highlighting future directions in cognitively bounded persona design.

Summary:
Your P-CBS approach—embedding mathematical penalties for both working memory decay/revisitation aversion and conditional uncertainty/cognitive load directly into a persona's state-space utility function during heuristic search—is novel relative to current published methods in automated playtesting/PCG evaluation literature.

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